

Dhaka Earthquake Helper

Project Report

1. Project Title and Member Information

Project Title: Dhaka Earthquake Helper

Institution: American International University-Bangladesh (AIUB)

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2. Problem Statement and Objectives

Problem Statement: During and immediately after an earthquake in a densely populated city like Dhaka, chaos ensues. Citizens often struggle to find which nearby hospitals are functional and have available beds, or where the designated safe open grounds are located. This lack of centralized, easily accessible information can lead to delayed medical attention and increased panic.

Objectives: The primary objective of this project is to provide a unified desktop application that can quickly locate the nearest hospitals with available capacity based on the user's location. Additionally, it aims to manage and display safe assembly grounds, maintain an emergency checklist for preparedness, and log search history to track emergency patterns and severity.

3. Feature List and Target Users

Target Users: Residents of Dhaka, local volunteers, and students who need a quick, offline desktop reference to find hospital capacities and safe assembly grounds during an emergency.

Detailed Feature List:

- **Hospital Locator System (tab_locate.py)**
 - **Emergency Input Form:** Users can select their location from a dropdown and input their age, gender, injury severity (Minor, Moderate, Severe), and toggle a Priority checkbox.
 - **Distance Calculation:** Uses NumPy (np.linalg.norm) to calculate the exact Euclidean distance between the user's selected location coordinates and the hospital coordinates.
 - **Top 3 Recommendations:** Displays the three closest hospitals, converting the coordinate distance into an estimated kilometer value.
 - **Capacity & Priority Logic:** Calculates the current hospital bed capacity percentage ($\text{patient_count} / \text{max_capacity} * 100$). If a user is marked as a Priority and the closest hospital is at 90% capacity or higher, the system actively flags it as "(Not Recommended - Full)".
- **Safe Grounds Management (tab_safegrounds.py)**
 - **Search and View:** Displays safe open spaces (like parks and stadiums) in a Tkinter Treeview table, allowing users to filter the list by typing a name or location.
 - **Export Data:** Includes an export button that formats the current safe grounds list and saves it locally as a standard .txt file for easy sharing.
 - **Data Entry & Deletion:** A management form allows adding new safe grounds. It includes validation to prevent negative capacities and uses a Python set to block duplicate IDs. Users can also select rows in the table to delete them from the JSON file.
- **Preparedness Guideline & Checklist (tab_guideline.py)**
 - **Safety Information:** Displays a read-only text box outlining standard procedures for before, during, and after an earthquake.
 - **Interactive Checklist:** Provides Checkbuttons for essential items (Water, First Aid Kit, Flashlight, Emergency Food).

- **Save & Statistics:** Progress is saved to a local checklist.txt file. Clicking "Show Statistics" uses NumPy (np.sum, np.mean) to calculate and display a popup containing the exact percentage of items prepared.
- **Query Logging & History (tab_history.py)**
 - **Automated Logging:** Every time a user clicks "Locate Nearest Help", the inputs and recommended hospital are silently saved to query_log.txt.
 - **Search & Filter:** Users can view past queries in a listbox and filter them using dropdowns for specific injury severities or priority statuses.
 - **Summary Statistics:** Calculates and displays a mathematical summary of the session, including total queries, average age, youngest/oldest patients, and the percentage of priority cases.
 - **CSV/TXT Export:** Allows the user to export the full query history and statistical summary into a cleanly formatted .csv or .txt file using Python's built-in csv module.

4. Explanation of Used Python Concepts (Week 1-5)

The project heavily relies on basic Python concepts to process data and make the application function correctly. Based on the project requirements, here is how we used them:

- **Variables and Data Types:** We used different types of variables to store the app's core information. We used text (strings) to store hospital names, safe ground IDs, and category choices (like "Severe" or "Minor"). We used numbers (integers and floats) to hold quantities like patient counts and calculated results like the distance in kilometers. We also used True/False values (booleans) to store status values, such as whether a user checked the priority box.
- **Operators:** We used math, comparison, and logical operators throughout the code. Math operators were used to perform arithmetic, such as calculating totals, finding the average age of users, and calculating the exact percentage of available hospital beds. Comparison and logical operators (like <, >, and, not) were used for validation checks, like making sure an age is between 1 and 120, and to make logical decisions about which hospital is best.
- **Branching (if/elif/else):** We used if, elif, and else statements to make decisions in the code and handle validation. For example, we use branching to check if a user left a field blank and show an error message if they did. We also use it for status decisions, like checking if a hospital's capacity is over 90% to flag it as full.
- **Loops (for/while):** for loops are used to process records and traverse lists and dictionaries. We use a loop to go through every single hospital in our JSON data to calculate which one is the closest. We also use loops to traverse the dictionary of safe grounds to show them in the UI. Finally, root.mainloop() is used to keep the application active and running until the user decides to exit.
- **Functions:** To keep the code organized, we created separate functions and methods for specific tasks. We built custom functions for input validation, calculation (get_capacity_percent), search (get_nearest_hospitals), saving and loading data (load_locations, save_data), updating and deleting records (delete_ground), and creating statistics (get_statistics).

5. Data Structure Explanation

- **Lists:** Used to keep a collection of items in order. For example, `self.hospitals = []` is an empty list that we fill with hospital objects. We also used lists to store the user's search history.
- **Tuples:** Used to group fixed items together so they cannot be changed. When finding the closest hospital, we put the distance, an index number, and the hospital object into a tuple like this: `(dist, i, h)`. This made it easy to sort them from closest to farthest.
- **Sets:** Used to store items where we don't want any duplicates. In the safe grounds code, we used `self.used_ids = set()` to remember which IDs are already taken. This makes sure no two safe grounds accidentally get the same ID.
- **Dictionaries:** Used to store information in "key-value" pairs for very fast searching. In our `locations.json` file, the city name (like "Mirpur") is the key, and its map coordinates are the value. We also used dictionaries to easily count how many "Severe" or "Minor" injury searches happened.

6. OOP Design: Class Name, Attributes, and Methods

The application uses Object-Oriented Programming (OOP) to organize the code into real-world concepts (Classes). Here are the main ones we built:

- **Class: Hospital**
 - **Attributes:** name, type, location, patient_count, max_capacity, coords
 - **Methods:** `get_capacity_percent()` (calculates how full it is), `display_info()`
- **Class: SafeGroundManager**
 - **Attributes:** filename, grounds (a dictionary), used_ids (a set)
 - **Methods:** `load_data()`, `save_data()`, `add_ground()`, `delete_ground()` (handles the data entry)
- **Class: EmergencyLocator**
 - **Attributes:** hospitals, locations
 - **Methods:** `get_nearest_hospitals()`, `get_statistics()` (does the math for distances and history)
- **Class: DataStore**
 - **Attributes:** data_file, log_file, log
 - **Methods:** `add_log_entry()`, `export_to_csv()`, `clear_log()` (manages saving and loading text files)
- **Class: Guideline**
 - **Attributes:** file, items
 - **Methods:** `add_item()`, `save()`, `load()`, `statistics()` (manages the emergency checklist)

7. File Handling and Exception Handling

- **File Handling:** The app needs to read and save data so nothing is lost when it closes. We used the `with open()` command to read and write our JSON files and text files (like `cases.txt` and `query_log.txt`). Using `with open()` is safe because it automatically

closes the file when it is done. We also used Python's built-in `csv` module to write the history summary into a clean Excel-friendly format.

- **Exception Handling:** We used `try...except` blocks to stop the app from crashing if something goes wrong.

- We catch `FileNotFoundError` and `JSONDecodeError` just in case the data files are missing or broken when the app starts.
- We catch `ValueError` in the forms if a user types something wrong (like typing letters instead of a number for their capacity) or if they try to add a Safe Ground ID that already exists.

8. Library/Module Use

- Tkinter & ttk: Serves as the primary graphical user interface framework, utilizing Notebooks for tabs, Treeviews for data display, and Messageboxes for alerts.
- NumPy (numpy): Crucial for mathematical operations in `locator.py`. Specifically, `np.array` and `np.linalg.norm` are used to calculate the precise Euclidean distance between the user's coordinates and hospital locations.
- JSON & CSV: Built-in libraries used to serialize/deserialize data and export structured emergency history logs for external analysis.
- OS: Used for path management and checking file existence before attempting to read logs.

9. Screenshots and Sample Input/Output

Sample Input/Output (Locate Tab):

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

Locate | Guideline | Find Safe Grounds | Manage Grounds | History & Stats

Enter Emergency Details

Location:

Age:

Gender:

Injury Severity:

☐ Priority (Woman/Children/Severe Injury)

Locate Nearest Help

Emergency Recommendations

Enter your details on the left and click 'Locate Nearest Help'...

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

Enter Emergency Details

Location:

Badda

Age:

21

Gender:

Male

Injury Severity:

Moderate

☐ Priority (Woman/Children/Severe Injury)

Locate Nearest Help

Emergency Recommendations

Your Location: Badda

Top 3 Nearest Hospitals:

1. United Hospital Limited

Distance: ~3.2 km

Type: Private | Capacity: 83%

2. Evercare Hospital Dhaka

Distance: ~3.7 km

Type: Private | Capacity: 70%

3. Holy Family Red Crescent Medical College

Distance: ~4.7 km

Type: Private | Capacity: 88%

Sample Input/Output (Guideline Tab):

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

Earthquake Guideline

BEFORE EARTHQUAKE

- Prepare emergency kit
- Keep important documents safe
- Save emergency contact numbers

DURING EARTHQUAKE

- Drop, Cover and Hold
- Stay away from windows
- Do not use elevators

☒ Water

☒ First Aid Kit

☒ Flashlight

☒ Emergency Food

Save Checklist

Show Statistics

Emergency Checklist

Done

Checklist Saved

OK

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LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

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- Prepare emergency kit
- Keep important documents safe
- Save emergency contact numbers

DURING EARTHQUAKE

- Drop, Cover and Hold
- Stay away from windows
- Do not use elevators

☐ Water

☐ First Aid Kit

☐ Flashlight

☐ Emergency Food

Save Checklist

Show Statistics

Sample Input/Output (Find Safeground Tab):

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

Search (Name or Location):

Search

Reset

Export List

ID	Name	Location	Capacity
1	Ramna Park	Shahbagh	8000
2	Suhrawardy Udyan	Shahbagh	12000
3	Chandrima Udyan	Mohammadpur	6000
4	Dhanmondi Lake Park	Dhanmondi	5000
5	Gulshan Lake Park	Gulshan	4000
6	National Parade Ground	Cantonment	20000
7	Osmani Udyan	Motijheel	3500
8	Hatigheel Open Area	Rampura	10000
9	Mirpur Cricket Stadium Ground	Mirpur	15000
10	Uttara Sector 4 Park	Uttara	3000

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

Search (Name or Location): Uttara

Search

Reset

Export List

ID	Name	Location	Capacity
10	Uttara Sector 4 Park	Uttara	3000

Sample Input/Output (Manage Grounds Tab):

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

LocateGuidelineFind Safe GroundsManage GroundsHistory & Stats

ID:

Name:

Location:

Capacity:

Add Ground

Delete Selected

ID	Name	Location	Capacity
1	Ramna Park		8000
2	Suhrawardy Udyan		12000
3	Chandrima Udyan		6000
4	Dhanmondi Lake Park		5000
5	Gulshan Lake Park		4000
6	National Parade Ground		20000
7	Osmani Udyan		3500
8	Hatigheel Open Area	Rampura	10000
9	Mirpur Cricket Stadium Ground	Mirpur	15000
10	Uttara Sector 4 Park	Uttara	3000
11	Badda School Ground	Badda	1000

Success

Safe ground added!

OK

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

Locate

Guideline

Find Safe Grounds

Manage Grounds

History & Stats

ID:

Name:

Location:

Capacity:

Add Ground

Delete Selected

	ID	Name	Location	Capacity
1		Ramna Park	Shahbagh	8000
2		Suhrawardy Udyan	Shahbagh	12000
3		Chandrima Udyan	Mohammadpur	6000
4		Dhanmondi Lake Park	Dhanmondi	5000
5		Gulshan Lake Park	Gulshan	4000
6		National Parade Ground	Cantonment	20000
7		Osmani Udyan	Motijheel	3500
8		Hatigheel Open Area	Rampura	10000
9		Mirpur Cricket Stadium Ground	Mirpur	15000
10		Uttara Sector 4 Park	Uttara	3000
11		Badda School Ground	Badda	1000

Dhaka Earthquake Helper

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After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

Locate

Guideline

Find Safe Grounds

Manage Grounds

History & Stats

ID:

Name:

Location:

Capacity:

Add Ground

Delete Selected

	ID	Name	Location	Capacity
1		Ramna Park	Shahbagh	8000
2		Suhrawardy Udyan	Shahbagh	12000
3		Chandrima Udyan	Mohammadpur	6000
4		Dhanmondi Lake Park	Dhanmondi	5000
5		Gulshan Lake Park	Gulshan	4000
6		National Parade Ground	Cantonment	20000
7		Osmani Udyan	Motijheel	3500
8		Hatigheel Open Area	Rampura	10000
9		Mirpur Cricket Stadium Ground	Mirpur	15000
10		Uttara Sector 4 Park	Uttara	3000

Success

Safe ground deleted successfully!

OK

Sample Input/Output (History and Stats Tab):

Dhaka Earthquake Helper

Dhaka Earthquake Helper

After an earthquake, this tool shows which nearby hospitals have space, recommends the best one for your situation, and points you to the nearest open ground for safe assembly and gives various guidelines and history.

Locate

Guideline

Find Safe Grounds

Manage Grounds

History & Stats

Past Emergency Searches

Search Location: Badda

Severity: All

Priority: All

Clear History

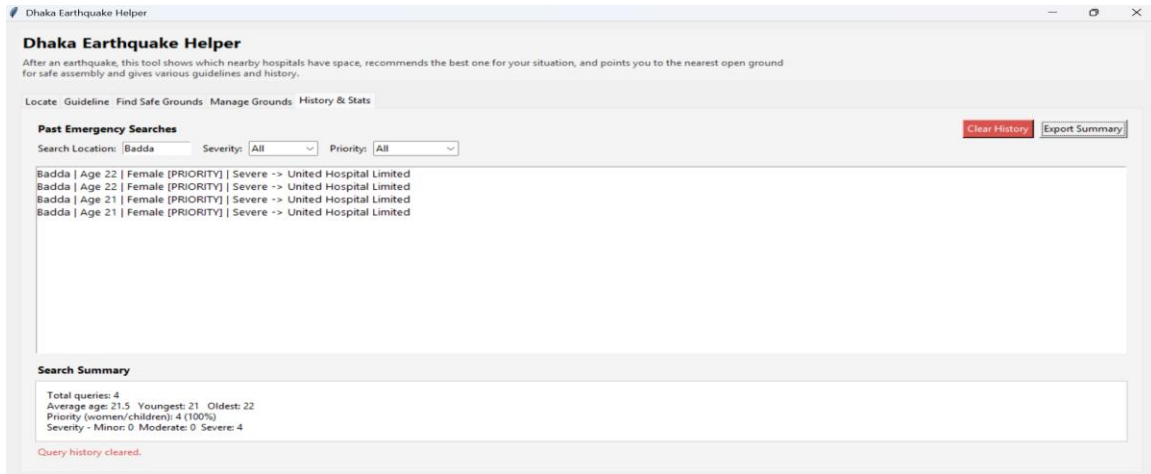
Export Summary

No queries logged yet.

Search Summary

No searches yet. Run a Locate search to create history and statistics.

Query history cleared.



10. Limitations and Possible Future Improvements

Limitations:

- The application currently relies on static JSON data for hospital capacities, which may not reflect real-time post-earthquake realities.
- Distance is calculated using Euclidean (straight-line) metrics via NumPy, which does not account for actual road networks or traffic blockages.
- Location nodes are limited to predefined city areas rather than dynamic GPS coordinates.

Future Improvements:

- API Integration: Connect to a live mapping service (like Google Maps API) for accurate routing and real-time GPS location tracking.
- Web Scraping / Database: Implement a backend server to fetch live hospital bed availability.
- Mobile Port: Rebuild the interface using a framework like Kivy or Flutter, as a desktop application is less practical during an active evacuation scenario.